



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

Can we trust the result?

Publication bias, robustness and sensitivity

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SOUTIENS INSTITUTIONNELS



Learning objectives

By the end of this session, you will be able to:

- 1 List the ways a set of studies can be a **biased sample** of the evidence.
- 2 Read a **contour-enhanced funnel plot** — and explain why asymmetry $\neq$ publication bias.
- 3 Test for small-study effects and time-lag bias **with dependent effect sizes**, and know which classic methods to avoid.
- 4 Run **sensitivity analyses** (leave-one-study-out, influence, study quality) and report robustness.

Are our 28 studies a fair sample?

A meta-analysis assumes the studies it contains are **representative** of all studies done.

But studies with **significant** results are more likely to be:

- **published** → *publication bias*
- published **quickly** → *time-lag bias*
- published **in English** → *language bias*
- published **more than once** → *multiple-publication bias*
- **cited** (and found) → *citation bias*

The consequence

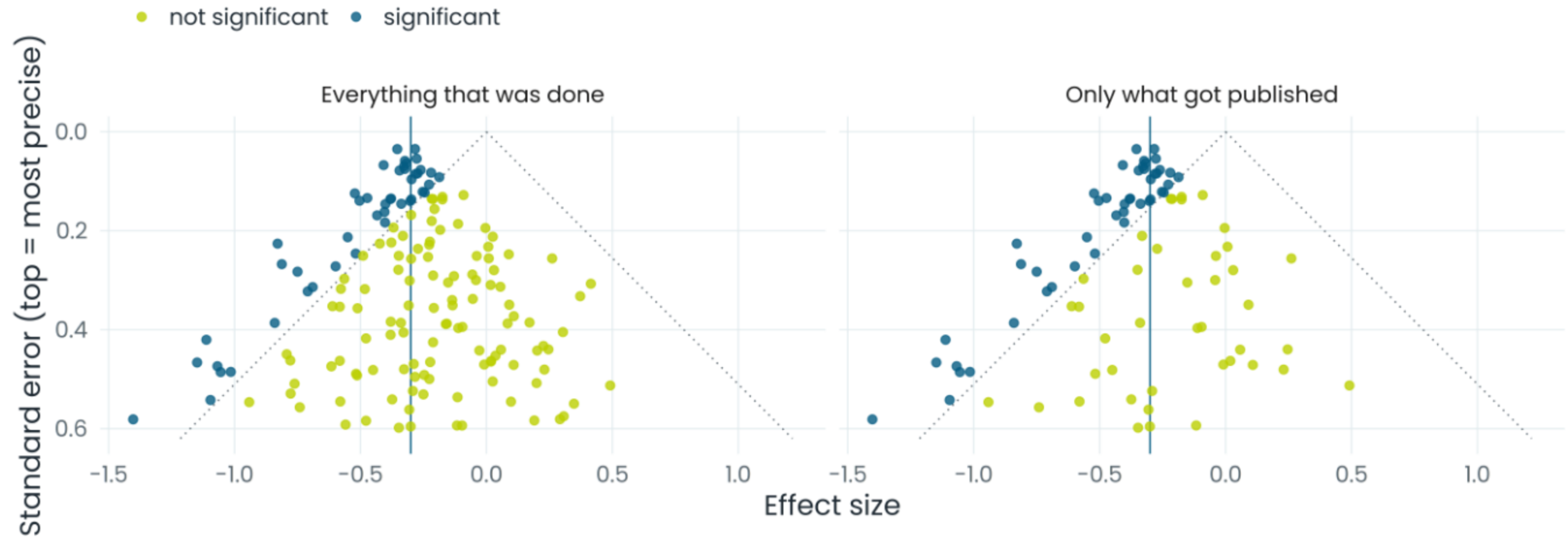
The pooled effect can be **overestimated** — sometimes an effect appears where there is none.

The best cure is prevention

Search grey literature, theses, reports and non-English studies (Monday's sessions). The tests we see today can only **detect** and **probe** bias — never repair it.

Publication bias and small-study effects

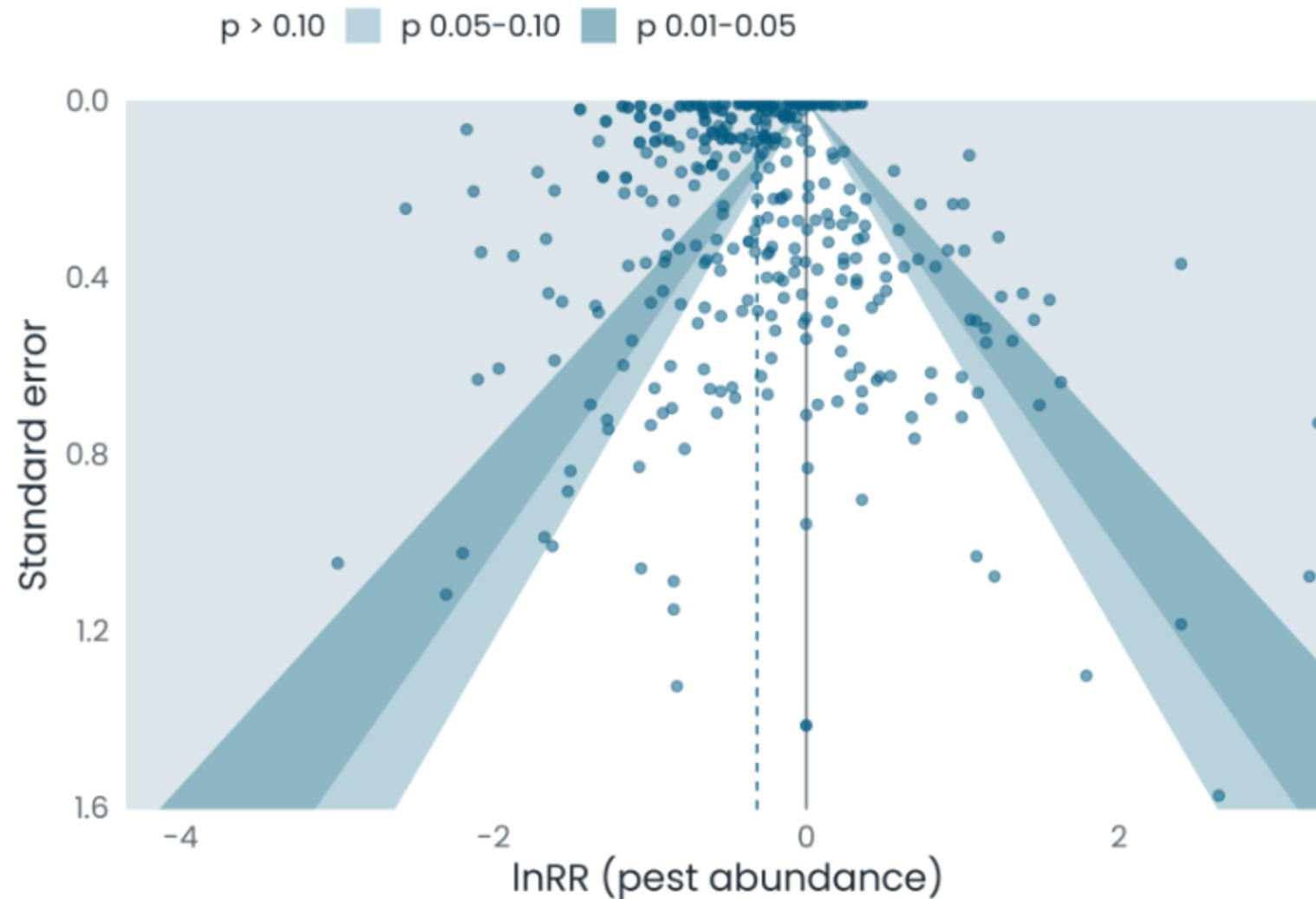
The logic of the funnel plot



Large studies (top) cluster near the true effect; small ones (bottom) scatter **symmetrically** → a funnel.

Where would the missing studies be?
Small **non-significant** studies (between the dotted lines) stay in drawers → a corner is **missing**.

Our funnel, with significance contours



- Shaded bands: significance of each comparison (grey background: $p < 0.01$)
- Dashed line: the multilevel mean
- Top: 83 comparisons from 2 studies with $SE < 0.02$ — suspiciously precise: check the extraction

If missing studies were in the **white** zone (non-significant), publication bias is a plausible cause. If they would sit in significant zones, look for other causes (Peters et al. 2008).

Asymmetry $\neq$ publication bias

YOUR CALL

Our funnel is asymmetric. List other reasons than publication bias.

- **Heterogeneity:** small studies differ (other crops, pests, methods) — true small-study effects.
- **Artefacts:** the variance of $\ln RR$ depends on the mean; extraction errors; pseudo-replication.
- **Chance:** with few studies, funnels are often asymmetric.

Testing for small-study effects: the classic way

Egger's test (Egger et al. 1997): regress each effect size on its standard error. A slope $\neq 0$ means small studies give different results.

It is a test of **funnel-plot asymmetry (small-study effects)**, not a test of publication bias: asymmetry can also come from heterogeneity or from design differences between small and large studies.

Our pests: $p = 0.00037$.

Two problems

1. It treats our 382 comparisons as **independent** — they come from 28 studies.
2. For lnRR, the SE is computed **from the means themselves** → an artificial link between effect and SE, even without any bias.

Testing for small-study effects: the multilevel way

Add a precision measure as a **moderator** of the multilevel model (Nakagawa et al. 2022), based on **sample size only**:

$$\sqrt{1/n_T + 1/n_C}$$

```
1 rma.mv(yi, vi,  
2       mods = ~ sqrt(1/n_t + 1/n_c),  
3       random = ~ 1 | study_id / es_id,  
4       test = "t", dfs = "contain",  
5       data = dat)
```

Precision measure	Slope p
SE (artefact-prone)	0.018
$\sqrt{1/n_T + 1/n_C}$	0.10
same, cluster-robust (CR2)	0.18

With the recommended measure: **no robust evidence** of a small-study effect.

If there were bias, would the conclusion change?

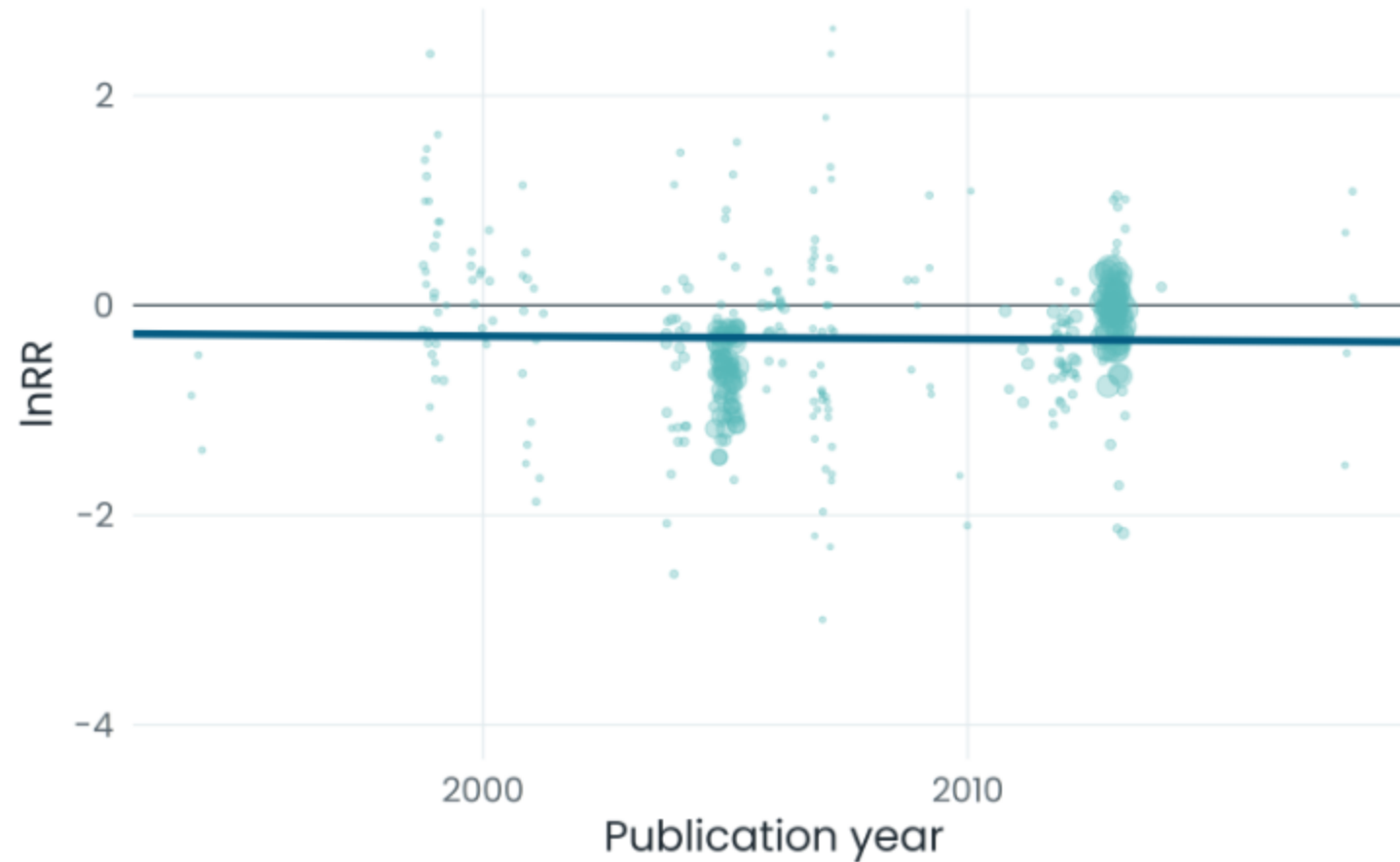
Estimate	Pest abundance [95% CI]
Main model	-27% [-40; -11]
PET : effect of an infinitely large study	-43% [-61; -18]
PEESE	-36% [-50; -16]

Rule (Stanley and Doucouliagos 2014): if the PET intercept is significant (here yes), report **PEESE**.

Adjusted estimates are, if anything, **stronger**: the pest reduction is **not** an artefact of small-study effects (whatever their cause).

With strong heterogeneity, PET-PEESE tends to **over-correct**: treat it as a sensitivity check, not as “the true effect”. Alternative: **selection models** [Vevea and Hedges (1995); `selmodel()` in `metafor`], which model the probability of publication given the p-value.

Time-lag bias



Do early studies report stronger effects than later ones?

Year as a moderator in the multilevel model: slope = -0.0029 per year ($p = 0.87$).

No evidence of a decline effect — with only 28 studies, the test has little power.

Also possible: include year and SE together (Nakagawa et al. 2022).

Two classics to avoid (or to use with care)

Fail-safe N (Rosenthal)

“How many null studies would make the result non-significant?”

Our data: **4,098,124** studies.

Meaningless: assumes missing studies have zero effect, ignores heterogeneity and dependence, estimates no bias. **Do not report it.**

Trim-and-fill

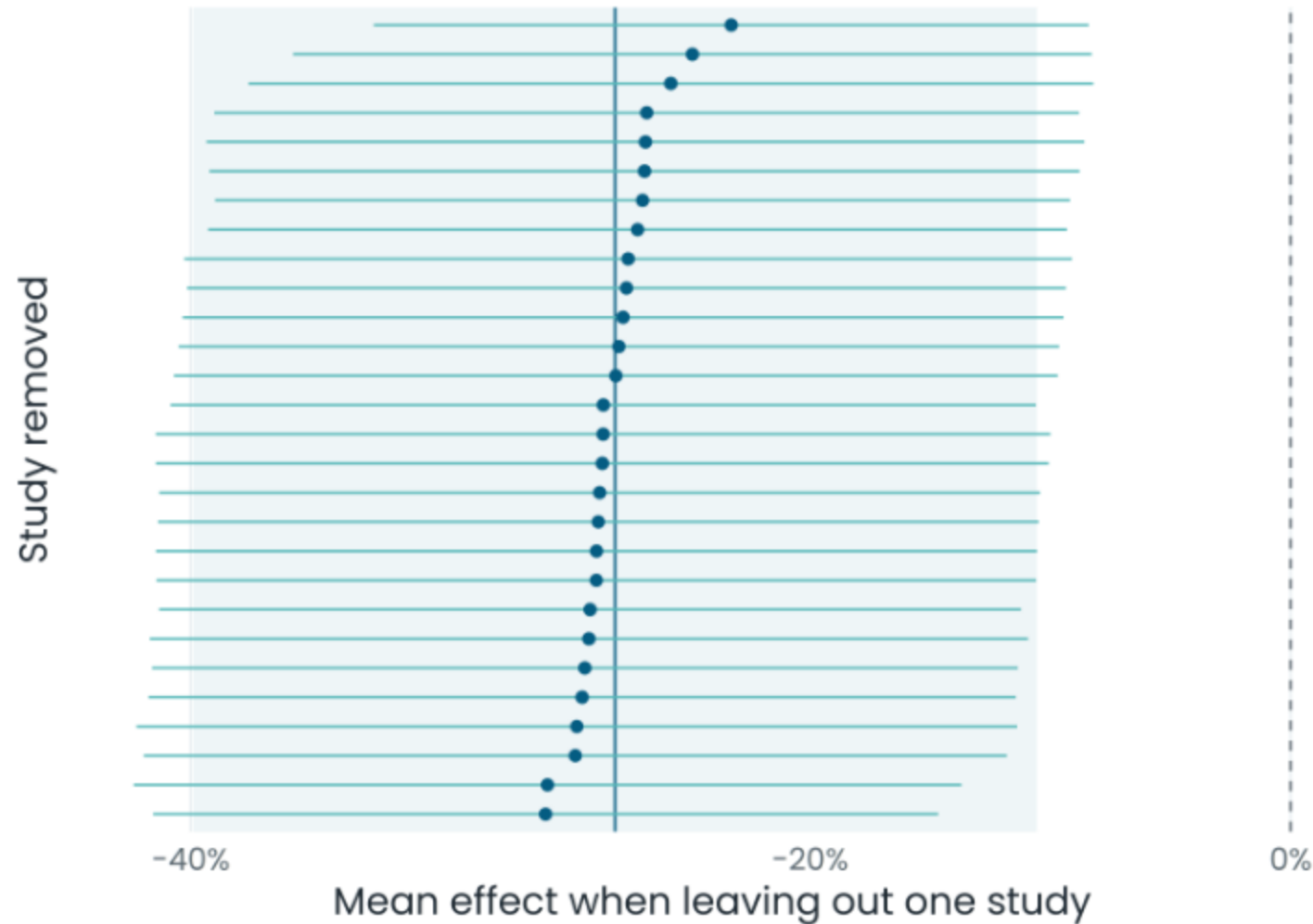
Imputes “missing” studies to make the funnel symmetric.

Assumes asymmetry = publication bias, performs badly with heterogeneity, ignores dependence.

→ at most a **sensitivity analysis** on study-level data, never “the corrected effect”.

Robustness

Does one study drive the result?



Leave-one-study-out: refit the model 28 times, dropping one whole study each time.

- Mean ranges from **-29%** to **-23%**
- The CI never crosses 0

Cook's distance by study: largest = 0.32, no study stands out from the others.

The conclusion does not depend on any single study.

A robustness table for your paper

Analysis	Pest abundance	Conclusion changes?
Main multilevel model (28 studies)	-27% [-40; -11]	—
Only comparisons meeting all validity criteria† (13 studies; 130 comparisons not appraisable)	-27% [-39; -13]; validity as moderator $p = 0.74$	No
Leave-one-study-out (range of means)	-29% to -23%	No
Small-study effect (sample-size based)	slope $p = 0.10$; PEESE -36%	No
Time-lag (year as moderator)	slope ≈ 0 ($p = 0.87$)	No (low power)

† Critical appraisal coded by Jones et al. (2021): enough sampling units in both groups, plots in their state long enough, intervention and comparator < 1.1 km apart. This is **internal validity** of each study (risk of bias, cf. CEE critical appraisal) — a different question from publication bias.

Also worth testing: another effect-size metric, the imputation of missing SDs, the correlation ρ assumed within studies. Report **all** sensitivity analyses — including those that change the result.

Practical · 33 min + debrief

Time	Task
15:22	Contour-enhanced funnel plot (code given: interpret)
15:27	Classic vs multilevel Egger (the lnRR trap); PET-PEESE
15:37	Sensitivity : leave-one-study-out, validity
15:47	<i>Your turn</i> : natural enemies – or the optional parts (time-lag, Cook, fail-safe N)
15:55	Debrief (all together)

Open **meta_analysis_course.Rproj** (the course kit), then **td_bias.R**. Instructions and folded solutions: the practical web page.

KEY MESSAGES

- 1 | Bias is best **prevented** by the search, not corrected by statistics.
- 2 | **Asymmetry $\neq$ publication bias**: read the direction, use contour-enhanced funnels.
- 3 | With several effect sizes per study, use **multilevel** tests — for lnRR with a **sample-size-based** precision; drop fail-safe N.
- 4 | Report a **robustness table**: leave-one-study-out, influence, study quality, analytical choices.

In the online notebook

Each point of this session developed further, with the same data, more pitfalls and exercises with solutions:

- Publication bias and sensitivity
- Data extraction and critical appraisal
- Reading a meta-analysis critically

literaturesynthesis.github.io/notebook

References

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